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Assignment Research: Report 3 Vulnerability Scanning with Shields Up and Nessus

What did you do?

I attempted to obtain permission from an organization to run the software scans, but was unsuccessful. I performed two complementary vulnerability scans on my home network. First, I used Shields UP!, a free browser-based tool from Gibson Research Corporation. This service remotely probes my public IP address and reports how my firewall and router respond, thereby revealing whether any TCP ports are exposed to the Internet. Because the site is stateless and relies solely on static HTML, it retains no personal data. After clicking Proceed, I ran the Common Ports test, followed by All Service Ports. Each results page highlights ports that are Open, Closed, or Stealth and provides a plain-text summary. I saved PDFs of the summaries for both scans and have included them in the results below.

For the second scan, I employed Nessus Essentials, Tenable's free entry-level vulnerability-assessment scanner licensed for up to sixteen unique IP addresses. After installation and activation, I launched a Host Discovery Scan to locate live devices and then ran a Basic Network Scan. Nessus operates by executing plug-ins written in NASL that perform banner grabs, configuration checks, patch-level verification, web-application fingerprinting, and more. It reports Open, Closed, and Filtered ports, and scores each finding with CVSS. When the scan completed, I reviewed the dashboard and noted the severity of each finding. Nessus supplies

remediation guidance and links to additional resources. I applied the recommended fixes until only one Medium finding remained. PDFs and screenshots of both scans accompany this report.

What are the results?

Screen shots from the scans appear below.

GRC Port Authority Report created on UTC: 2025-07-31 at 16:41:35

Results from scan of ports: 0, 21-23, 25, 79, 80, 110, 113,  
119, 135, 139, 143, 389, 443, 445,  
1002, 1024-1030, 1720, 5000

```

    0 Ports Open
    11 Ports Closed
    15 Ports Stealth
-----
    26 Ports Tested

```

NO PORTS were found to be OPEN.

Ports found to be CLOSED were: 0, 1002, 1024, 1025, 1026, 1027,  
1028, 1029, 1030, 1720, 5000

Other than what is listed above, all ports are STEALTH.

TruStealth: FAILED - NOT all tested ports were STEALTH,  
- NO unsolicited packets were received,  
- A PING REPLY (ICMP Echo) WAS RECEIVED.

The 26 Common Ports tested:

- 0 Ports Open.
- 11 Ports Closed.
- 15 Ports Stealth.
- TruStealth: FAILED.

GRC Port Authority Report created on UTC: 2025-07-31 at 16:38:51

Results from scan of ports: 0-1055

0 Ports Open  
72 Ports Closed  
984 Ports Stealth

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1056 Ports Tested

NO PORTS were found to be OPEN.

Ports found to be CLOSED were: 0, 1, 2, 3, 4, 5, 6, 30, 60,  
61, 91, 92, 121, 122, 151, 152,  
181, 182, 211, 212, 241, 242,  
271, 272, 299, 300, 329, 330,  
360, 361, 390, 391, 420, 421,  
450, 451, 480, 481, 512, 543,  
573, 574, 603, 604, 633, 634,  
663, 664, 693, 695, 724, 725,  
754, 755, 784, 785, 815, 816,  
844, 846, 875, 876, 905, 906,  
935, 936, 966, 967, 995, 997,  
1026, 1027

Other than what is listed above, all ports are STEALTH.

TruStealth: FAILED - NOT all tested ports were STEALTH,  
- NO unsolicited packets were received,  
- A PING REPLY (ICMP Echo) WAS RECEIVED.

The 1056 All Service Ports tested:

- 0 Ports Open.
- 72 Ports Closed.
- 984 Ports Stealth.
- TruStealth: FAILED.



#### Vulnerabilities by Host:

- 0 Critical.
- 0 High.
- 5 Medium: SMB Signing not required, 2 SSL Certificates Cannot Be Trusted, SSL Certificate Expired, and SSL Self-Signed Certificate.
- 0 Low.
- 112 Info (Risk Factor: None).

## Vulnerabilities by Host

[Collapse All](#) | [Expand All](#)

192.168.███



## Scan Information

Start time: Sat Aug 2 21:42:55 2025  
End time: Sat Aug 2 22:00:12 2025

## Host Information

Netbios Name: █████  
IP: 192.168.███  
OS: Windows 11

## Vulnerabilities

51192 - SSL Certificate Cannot Be Trusted

## Synopsis

The SSL certificate for this service cannot be trusted.

## Description

The server's X.509 certificate cannot be trusted. This situation can occur in three different ways, in which the chain of trust can be broken, as stated below :

## Vulnerabilities by Host After Remediation:

- 0 Critical.
- 0 High.
- 1 Medium: 1 SSL Certificate Cannot Be Trusted.
- 0 Low.
- 88 Info (Risk Factor: None).

What did you learn?

I learned that a scan in either Shields UP! or Nessus Essentials is considered successful only when no vulnerabilities are found. The Shields UP! Common Ports scan immediately pinpoints the ports most likely to be exploited by a malicious actor. The All Service Ports scan reveals obscure services, misconfigured firewall or router rules, and ISP filtering. Of the three possible states, Stealth status is preferred because packets are silently dropped (there is no response), so the port appears invisible. A Closed port indicates nothing is listening, yet it still replies, making its presence known. An Open port means a service completes the TCP handshake and therefore accepts connections.

After reviewing my results, I discovered that my ISP already blocks certain ports, which appeared as Stealth. I also found that ISPs allow users to change certain router rules depending on the ISP. I learned that Trojan ports can exist outside the common list. Passing TruStealth would require every probed port to be Stealth and all ICMP pings to be ignored. Achieving that goal may require tightening some firewall rules, disabling unnecessary services, or relocating certain services.

From the Nessus Essentials scan, I learned exactly what resides on my network -- live hosts, OS fingerprints, open ports, and the services bound to those ports. This revealed items I had not realized attackers could exploit. Nessus flags outdated programs with known vulnerabilities and reports on ports, though it labels them as informational findings rather than outright vulnerabilities. On my system, Nessus flagged "SMB signing not required" and several certificate issues.

The flagged port led me to an application on my computer that I had downloaded to test, but never used. I also discovered that localhost certificates needed to be installed in my web

browser's certificate store. Correcting configurations cleared all but one SSL warning on port 51192. Although Nessus allows you to save the certificate, it does not remove the flag because it trusts only certificates issued by publicly recognized Certificate Authorities. Nessus further checks for default or weak credentials and unsecured services or data. The report's comprehensive view provided findings for each device. Nessus groups devices by Family and counts them, enabling me to see which categories are most at risk. I was pleased that clear remediation resources were available, and now I feel that my home network is more secure.

In short, Nessus exposes the internal weaknesses, while Shields UP! reveals external exposure. Understanding both perspectives is crucial for a holistic view of cybersecurity posture. As I add new devices or software to my network, I can run Nessus scans to identify any known vulnerabilities, missing patches, or misconfigurations, thereby continuously improving security over time. New vulnerabilities are discovered every day, and developing the habit of scanning and remediating creates an ongoing cycle of security enhancement for my home network. Using Shields UP! periodically will verify that only intended ports remain open and everything else is Stealth. Over time, this practice will help ensure that no unintended exposure occurs, and I will gain practical experience and insight from each scan.